

Budgeted Planning and Deterministic Guards for CAR-bench Track 2

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Abstract

Our Track 2 agent obtains 36.0% official Pass³ and 54.7% Pass@3 on the full 125-task CAR-bench public test set (375 episodes). Its split Pass³ scores are 48% Base, 36% Hallucination, and 24% Disambiguation. The fixed gpt-oss-120b model operates behind an ordered deterministic controller chain, a risk-tiered planner-executor, and deterministic output guards. The submitted configuration has no parallel LLM fan-out and a reachable maximum sequential depth of three. The public audit confirms a maximum of three calls across 720 metered assistant steps; no step exceeds the limit of five. The main failures are systematic. On one fixed 36-task development slice, cumulative stages raise Pass³ from 41.7% to 83.3% while reducing maximum depth from 14 to 3. On public test, missing parameters obtain 0% Pass³, missing response fields obtain 9.1%, and internal Disambiguation obtains 7.7%. Call-budget compliance is therefore necessary but not sufficient.

1 Result and Contributions

CAR-bench tests tool use under normal operation, missing capabilities, and ambiguous requests [Kirmayr et al., 2026]. A task contributes to Pass³ only when all three trials pass. Pass@3 records whether at least one trial passes [Yao et al., 2024]. The difference measures unstable success.

Track 2 fixes the open-weight gpt-oss-120b model [OpenAI, 2025]. It limits sequential LLM calls to five per benchmark-visible assistant step and sets an average token budget. We address these constraints with three mechanisms: (1) deterministic controllers that can act without an LLM call; (2) a serial planner-executor path selected by visible risk signals; and (3) a bounded structured-output recovery path. We also report a complete public-test reliability and compute audit.

The frozen submission uses image digest sha256:fb0c8f6... and planner commit f2a0d77. It scored 178/375 public episodes. The official split-macro

Pass³ is 36.0%. This exceeds the published direct gpt-oss-120b reference of 28%, but the comparison is contextual because configurations can differ.

2 Architecture

Figure 1 shows one assistant step. The agent uses only the conversation, current tool schema, and returned tool results. It proposes tool calls across the A2A boundary; the evaluator executes them.

Deterministic fast path. An ordered set of controllers checks capability, preferences, required observations, state-changing commitments, relative-state requests, and bundled goals. The Grounded Subgoal Execution Graph (GSEG) records requested subgoals and observed completion. A matched controller can return a grounded response or proposed call at depth zero.

Budgeted model path. A rule-based estimator assigns low, medium, high, or critical risk. The allocated ceilings are 1, 2, 4, and 5 calls. A fresh medium-or-higher user turn uses a private planner. Tool-result continuations reuse the active plan. Low-risk steps use a deterministic fallback plan. The executor then produces one structured action. Planner and executor both use medium reasoning effort, with completion caps of 2048 and 1024 tokens.

Guards and recovery. Deterministic postprocessors normalize arguments and enforce schema, evidence, policy, and completion conditions. They do not call the model. High and critical tiers permit one retry only when the planner or executor returns malformed structured output. Thus, the submitted configuration can reach depth three even though the hard allocator ceiling is five.

3 Evaluation Protocol

We developed on train tasks, selected a champion with fixed promotion gates, and froze the image before the public run. The public evaluation contains 50 Base, 50 Hallucination, and 25 Disambiguation tasks with three trials each. Thirteen restartable chunks cover all 375 task-trial pairs. The selected artifacts contain no missing or duplicate pairs, empty trajectories, or infrastructure-error episodes.

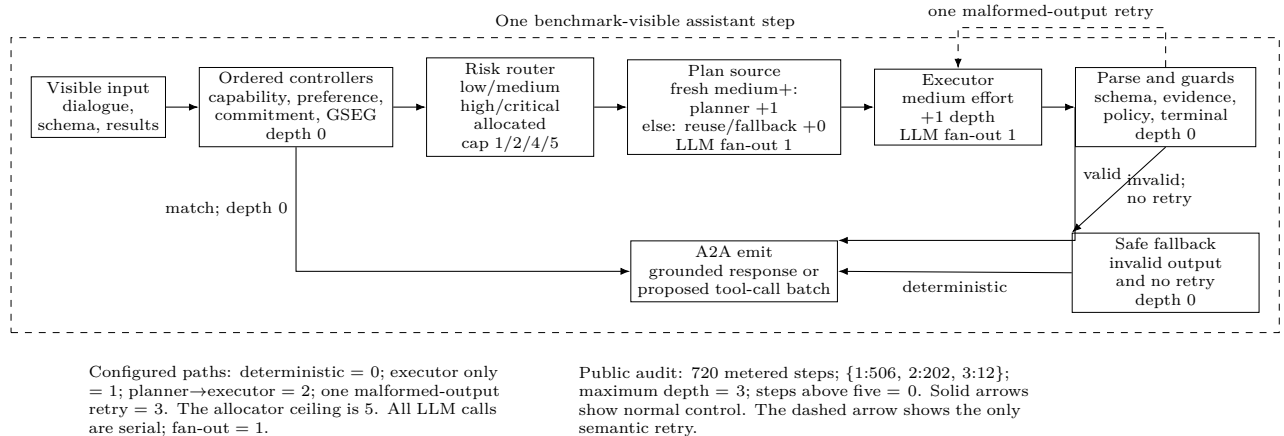


Figure 1: Track 2 architecture and per-step LLM-call depth. Deterministic postprocessors do not invoke an LLM repair. Provider transport retries are outside the reported application-level call metric.

Official overall metrics are unweighted means over the three task families. Raw episode accuracy is task-weighted. Development results below use curated train slices and do not estimate public-test performance.

4 Experiment Program

The research program starts from bounded rationality: spend model calls only when a cheap controller cannot decide safely [Simon, 1955]. STRIPS-style preconditions and affordance checks motivate capability and commitment controllers [Fikes and Nilsson, 1971; Ahn et al., 2022]. Belief-state control motivates evidence provenance and internal lookup before action [Kaelbling et al., 1998]. Failure audits then select the next mechanism. This is a cumulative controller program, not prompt search.

Each challenger first passed focused three-trial gates. It then faced the same balanced 36-task slice. Promoted candidates advanced to a 66-task gate and a matched head-to-head. The frozen image finally ran on all public tasks. We use $\mathcal{L}_{\text{rel}} = 1 - \text{Pass}^3$ as a descriptive reliability loss, not as a gradient-trained objective.

The initial controller stack reached 54/54 on an 18-task gate, then fell to 50.0% Pass^3 on the first 36-task run. This exposed development overfit. Commitment discipline produced the largest class-level change: Disambiguation raw score rose from 11/36 to 27/36. Relative-state derivation added one raw pass but introduced comfort-scope failures in 9 of 20 audited real failures. Scope and schema preservation repaired that side effect and raised the total from 86/108 to 91/108.

The wider 66-task audit then found 59 real failures after excluding five evaluator artifacts: incomplete bundles accounted for 18, route continuity for 17, contextual grounding for 13, and preference binding for 8. This distribution motivated GSEG. GSEG reached 92/108 on the 36-task gate. The submitted stabilization later reached 98/108. In a matched 66-task head-to-head it scored 139/198 and 63.6% Pass^3 , versus 132/198 and

Public test	Raw	P ¹	P ²	P ³	P@3
Base	85/150	56%	56%	48%	62%
Hallucination	65/150	40%	38%	36%	54%
Disambiguation	28/75	32%	28%	24%	48%
Official macro	178/375	42.7%	40.7%	36.0%	54.7%

Table 1: Full public-test results for the frozen Track 2 submission.

Family	Subtype	Tasks	Raw	P ³	P@3
Hall.	Missing tool	29	55/87	58.6%	72.4%
Hall.	Missing parameter	10	5/30	0.0%	30.0%
Hall.	Missing response field	11	5/33	9.1%	27.3%
Dis.	Internal evidence	13	6/39	7.7%	23.1%
Dis.	User-resolvable	12	22/36	41.7%	75.0%

Table 2: Public-test subtypes. The agent detects whole-tool removal but does not transfer this behavior to schema deltas or internal evidence.

57.6% for GSEG; Disambiguation tied at 27/66. We promoted only after this comparison.

5 Results

Table 1 is the primary result. The 18.7-point gap between Pass^3 and $\text{Pass}@3$ is smaller than Track 1’s gap, but for a worse reason: 55 tasks fail in all trials, 22 pass in one or two trials, and 48 pass in all trials. Most remaining errors need a new mechanism, not more sampling.

The fall from 63.6% on the 66-task train slice to 36.0% on public test is a descriptive 27.6-point generalization gap; the task sets differ. We use the public result as the reliability estimate.

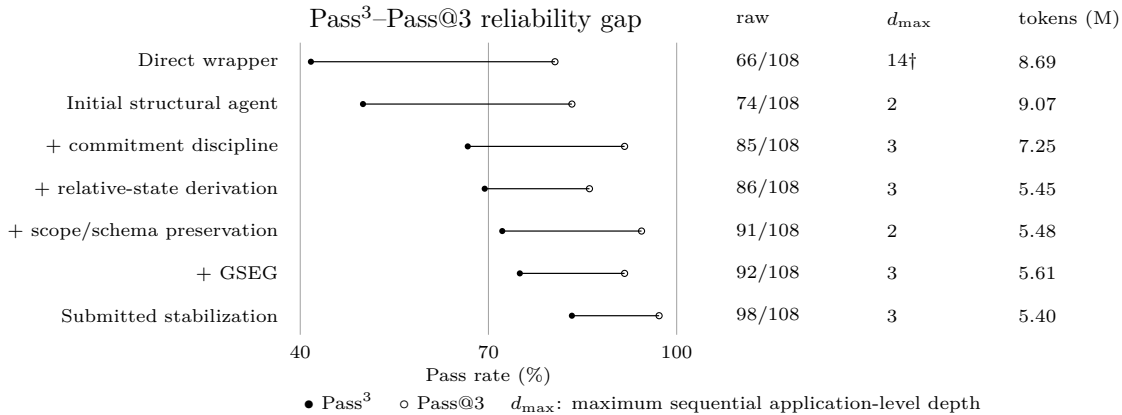


Figure 2: Track 2 staged development on one fixed 36-task slice (108 episodes per row). Rows are independent cumulative runs, not paired single-mechanism ablations. Reliability loss falls from 58.3% for the direct wrapper to 16.7% for the submission while the maximum depth falls from 14 to 3. The direct wrapper has 10 steps above the limit of five (†).

Public compute metric	Value
Metered assistant steps	720
Sequential-call histogram	{1:506, 2:202, 3:12}
Maximum / steps above five	3 / 0
Parallel LLM fan-out	1
Mean tokens per evaluated episode	71,150
Mean tokens per unique three-trial task	213,451
Median / 95th percentile tokens per episode	48,014 / 197,560
Maximum tokens / maximum A2A turns	1,134,081 / 40

Table 3: Compute audit of the 375 public episodes. Token counts include input, output, and reasoning tokens recorded by the agent.

6 Compute and Failure Analysis

Table 3 confirms sequential-call compliance. The average token use per unique task remains below the approximate 500k-token allowance. However, three failed episodes exceed 500k tokens. Long failed loops also reach 40 A2A turns. Progress detection is therefore a reliability mechanism, not only an efficiency optimization.

Post-freeze experiments also define useful negative results. A schema-law stack scored 91/108 against the 98/108 train champion. A reduced stack scored 87/108. A deterministic loop guard passed a small regression gate but changed one Disambiguation task from 3/3 to 0/3 because it suppressed a confirmed repeat action. We rejected all three changes. Future loop detection must distinguish user-confirmed repetition from stalled execution.

The full public test identifies two primary mechanism gaps. First, the capability model represents tool presence better than parameter or response availability. Second, internal evidence retrieval does not reliably bind preferences, contacts, calendar state, or context to action parameters. These gaps explain the low Hallucination and Disambiguation scores more directly than the call budget does. Hidden-test scores and ranking are not included.

7 Conclusion

The submitted Track 2 agent is compliant and compute-aware. It obtains 36.0% official Pass³ and uses at most three sequential application-level LLM calls in the public audit. Deterministic controllers and budgeted planning raise performance above the published direct-model reference. The remaining 55 systematic failures require schema-delta reasoning, evidence binding, and loop-progress invariants.

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